United States Patent Application Publication 20250273476
Kind Code A1
Publication Date August 28, 2025
Inventor(s) TAVAKOLI; Mohammad Mahdi et al.

METHODS OF FORMING LOW RESISTIVITY FILMS USING Microwave treatment

Abstract

According to one or more embodiments, a method includes exposing a semiconductor device structure to a microwave process to cause impurities within at least one electrical connection formed in at least one feature of the semiconductor device structure to rise to a surface of the at least one electrical connection, and exposing the semiconductor device structure to a reactive gas to remove the impurities from the surface of the at least one electrical connection.

Inventors: **TAVAKOLI; Mohammad Mahdi (Santa Clara, CA), SHIN; Yoon Ah (Santa Clara, CA), GELATOS; Avgerinos V. (Scotts Valley, CA), LEE; Joung Joo (San Jose, CA), MEBARKI; Bencherki (Santa Clara, CA)**

Applicant: **Applied Materials, Inc.** (Santa Clara, CA)

Family ID: **1000007723181**

Appl. No.: **18/590102**

Filed: **February 28, 2024**

Publication Classification

Int. Cl.: **H01L21/321** (20060101); **H01L21/67** (20060101); **H01L21/768** (20060101)

U.S. Cl.:

CPC **H01L21/321** (20130101); **H01L21/67017** (20130101); **H01L21/67115** (20130101); **H01L21/76883** (20130101);

Background/Summary

BACKGROUND

Field

[0001] Embodiments of the present disclosure generally relate to a method and apparatus for forming low resistivity electrical connections.

Description of the Related Art

[0002] Integrated circuits have evolved into complex devices that can include millions of transistors, capacitors, and resistors on a single chip. In the course of integrated circuit evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased.

[0003] Microelectronic devices are fabricated on a semiconductor substrate as integrated circuits in which various conductive layers are interconnected with one another to permit electronic signals to propagate within the device. Examples of such devices include memory (e.g., dynamic random access memory (DRAM)) and logic devices, including both planar and three-dimensional structures. Three-dimensional structures include fin field-effect transistor (finFET) or metal-oxide-semiconductor field-effect transistor (MOSFET) devices.

[0004] Traditional middle-end-of-the-line (MEOL) electrical connections are formed in a feature, also referred to as a via, a trench, or the like, in the semiconductor substrate. MEOL electrical connections allow connections between front-end-of-the-line (FEOL) semiconductor structures and back-end-of-the-line (BEOL) interconnects. Electrical connections with a low resistivity are desirable in semiconductor devices. However, when MEOL electrical connections have high resistance, the electrical connections produce poor connections between the FEOL structures and the BEOL interconnects, reducing the performance of the packaged semiconductor structures. For example, MEOL electrical connections may have a resistivity due to a high level of impurities present in the electrical connection (i.e., the conductive material deposited in the feature to form the electrical connection).

[0005] Typically to reduce the level of impurities in the electrical connection, the electrical connections are treated using a plasma process. However, the plasma process is a surface based method and is not able to penetrate (treat) the bulk of the conductive material. Thus, there is a need for improved methods to reduce electrical connections and simplified processes of contact formation.

SUMMARY

[0006] According to one or more embodiments a method includes exposing a semiconductor device structure to a microwave process to cause impurities within at least one electrical connection formed in at least one feature of the semiconductor device structure to rise to a surface of the at least one electrical connection, and exposing the semiconductor device structure to a reactive gas to remove the impurities from the surface of the at least one electrical connection.

[0007] According to one or more embodiments, a processing tool includes a controller, and a memory for storing instructions, which, when executed by the controller, causes the controller to perform a method for treating an electrical connection formed in a feature of a semiconductor structure, the method comprising exposing a semiconductor device structure to a microwave process to cause impurities within at least one electrical connection formed in at least one feature of the semiconductor device structure to rise to a surface of the at least one electrical connection, and exposing the semiconductor device structure to a reactive gas to remove the impurities from the surface of the at least one electrical connection.

[0008] According to one or more embodiments, a method includes exposing a semiconductor device structure to a microwave process to cause impurities in an electrical connection comprising ruthenium (Ru) that is formed in a feature of the semiconductor device structure to rise to a surface of electrical connection, and exposing the semiconductor device structure to hydrogen to remove the impurities from the surface of the electrical connection.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments of the disclosure and are therefore not to be considered limiting of its scope, as the disclosure may admit to other equally effective embodiments.

[0010] FIG. **1** illustrates a schematic top view of a multi-chamber processing system, according to embodiments described herein.

[0011] FIG. **2** is a flow diagram depicting a method of treating an electrical connection of a semiconductor structure, according to one or more of the embodiments described herein.

[0012] FIGS. **3**A-**3**C illustrate views of various stages of treating an electrical connection of a semiconductor structure in accordance with one or more embodiments described herein.

[0013] FIGS. **4**A-**4**D, illustrate views of a microwave processing tool is shown, according to embodiments described herein.

[0014] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0015] Middle-end-of-the-line (MEOL) and back-end-of-the-line (BEOL) electrical connections, such as contacts, interconnects, and the like are formed by filling a feature such as a cavity, trench, or via with a conductive material. However, due to impurities within the conductive material, MEOL and BEOL electrical connections have a high resistivity hindering the performance of packaged semiconductor structures. Typically to reduce the amount of impurities in the electrical connection (conductive material), the electrical connections are treated using a plasma process. However, the plasma process is a surface based method and is not able to penetrate (treat) the bulk of the conductive material. Embodiments herein relate to treating the electrical connections with a microwave process that causes the impurities to rise to the surface of the electrical connection and then removing the impurities by exposing the semiconductor structures to a reactive gas.

Processing System Example

[0016] FIG. **1** illustrates a schematic representation of a processing system **100** for use with one or more embodiments of the disclosure. As detailed below, substrates in the processing system **100** may be processed in and transferred between the various chambers without exposing the substrates to an ambient environment exterior to the processing system **100** (for example, an atmospheric ambient environment such as may be present in a fab). For example, the substrates may be processed in and transferred between the various chambers maintained at a pressure (for example, less than or equal to about 500 Torr) or sub-atmospheric pressure, such as a vacuum environment, without breaking the reduced relative pressure or vacuum environment among various processes performed on the substrates in the processing system **100**. Accordingly, the processing system **100** may provide for an integrated solution for some processing of substrates.

[0017] Examples of a processing system that may be suitably modified in accordance with the teachings provided include the Endura®, Producer® or Centura® integrated processing systems or other suitable processing systems commercially available from Applied Materials, Inc., located in

Santa Clara, California (CA), United States of America. One may envision that other processing systems, including those from other manufacturers, may be adapted to benefit from aspects described.

[0018] FIG. **1** is a schematic top view of the substrate processing system **100** (also referred to as a “processing platform”), according to embodiments described herein. The processing system can be used for a microwave treatment of a surface of an electrical connection, such as a contact structure, an interconnect, or the like, to remove contaminants and impurities from the surface of the electrical connection. The substrate processing system **100** generally includes an equipment front-end module (EFEM) **102** for loading substrates into the processing system **100**, a first load lock chamber **104** and a second load lock chamber **106** coupled to the EFEM **102**, a transfer chamber **108** coupled to the first load lock chamber **104** and the second load lock chamber **106**, and a plurality of other chambers coupled to the transfer chamber **108** as described in detail below. The EFEM **102** generally includes one or more robots **105** that are configured to transfer substrates from front opening unified pods (FOUPs) **103** to at least one of the first load lock chamber **104** or the second load lock chamber **106**. Proceeding counterclockwise around the transfer chamber **108** from the buffer portion **108**A of the first load lock chamber **104**, the processing system **100** includes a first dedicated degas chamber **109**, a first pre-clean chamber **110**, a first pass-through chamber **112**, a second pass-through chamber **113**, a second pre-clean chamber **114**, a second degas chamber **116** and the second load lock chamber **106**. The buffer portion **108**A of the transfer chamber **108** includes a first robot **115** that is configured to transfer substrates to each of the load lock chambers **104**, **106**, the degas chambers **109**, **116**, the pre-clean chambers **110**, **114** and the pass-through chambers **112**, **113**.

[0019] The back-end portion **108**B of the transfer chamber **108** includes a second robot **135** that is configured to transfer substrates to each of the pass-through chambers **112**, **113** and the processing chambers coupled to the back-end portion **108**B of the processing system **100**. The processing chambers can include a first processing chamber **132**, a second processing chamber **134**, a third processing chamber **136**, a fourth processing chamber **138** and a fifth process chamber **140**. In general, the processing chambers **132**, **134**, **136**, **138**, **140** can include an atomic layer deposition (ALD) chamber, microwave chamber, chemical vapor deposition (CVD) chamber, physical vapor deposition (PVD) chamber, etch chamber, degas chamber, an anneal chamber, or other type of semiconductor substrate processing chamber. In some embodiments, one or more of the processing chambers **132**, **134**, **136**, **138**, **140** are a microwave chamber. In some examples, the processing chamber **110** may be capable of performing an etch process, the processing chamber **114** may be capable of performing a cleaning process or an annealing process, and the processing chambers **132**, **134**, **136**, **138**, **140** may be capable of performing respective microwave, CVD or ALD deposition processes. In one example, the processing chambers **132**, **134**, **136**, **138**, or **140** may be a Volta™ CVD/ALD chamber, or Encore™ PVD chambers available from Applied Materials of Santa Clara, Calif.

[0020] A system controller **126**, such as a programmable computer, is coupled to the processing system **100** for controlling one or more of the components therein. For example, the system controller **126** may control the operation of one or more of the processing chambers, such as processing chambers **132**, **134**, **136**, **138**, **140**. In operation, the system controller **126** enables data acquisition and feedback from the respective components to coordinate processing in the processing system **100**.

[0021] The system controller **126** includes a programmable central processing unit (CPU) **126**A, which is operable with a memory **126**B (for example, non-volatile memory) and support circuits **126**C. The support circuits **126**C (for example, cache, clock circuits, input/output subsystems, power supplies, etc., and combinations thereof) are conventionally coupled to the CPU **126**A and coupled to the various components within the processing system **100**.

[0022] In some embodiments, the CPU **126**A is one of any form of general purpose computer processor used in an industrial setting, such as a programmable logic controller (PLC), for controlling various monitoring system component and sub-processors. The memory **126**B, coupled to the CPU **126**A, is non-transitory and is typically one or more of readily available memory such as random access memory (RAM), read only memory (ROM), floppy disk drive, hard disk, or any other form of digital storage, local or remote.

[0023] Herein, the memory **126**B is in the form of a computer-readable storage media containing instructions (for example, non-volatile memory), that when executed by the CPU **126**A, facilitates the operation of the processing system **100**. The instructions in the memory **126**B are in the form of a program product such as a program that implements the methods of the present disclosure (for example, middleware application, equipment software application, etc.). The program code may conform to any one of a number of different programming languages. In one example, the disclosure may be implemented as a program product stored on computer-readable storage media for use with a computer system. The program(s) of the program product define functions of the embodiments (including the methods described herein). Illustrative computer-readable storage media include, but are not limited to: (i) non-writable storage media (for example, read-only memory devices within a computer such as CD-ROM disks readable by a CD-ROM drive, flash memory, ROM chips or any type of solid-state non-volatile semiconductor memory) on which information is permanently stored; and (ii) writable storage media (for example, floppy disks within a diskette drive or hard-disk drive or any type of solid-state random-access semiconductor memory) on which alterable information is stored. Such computer-readable storage media, when carrying computer-readable instructions that direct the functions of the methods described herein, are embodiments of the present disclosure. The various methods disclosed herein may generally be implemented under the control of the CPU **126**A by the CPU **126**A executing computer instruction code stored in the memory **126**B (or in memory of a particular processing chamber) as, for example, a software routine. When the computer instruction code is executed by the CPU **126**A, the CPU **126**A controls the chambers to perform processes in accordance with the various methods.

[0024] As will be described in more detail below, in one example of the processing system **100**, a semiconductor device structure undergoes a microwave process and an exposure to a reactive gas. In some embodiments, the microwave process and the exposure to the reactive gas are performed in different processing chambers. For example, the microwave process may be performed in the first processing chamber **132** and the exposure to the reactive gas may be performed in the second processing chamber **134**. In another example, the microwave process and the exposure to the reactive gas may be performed in a same processing chamber, such as the first processing chamber **132**.

Processing Sequence Example

[0025] FIG. **2** is a flow diagram depicting a method of treating an electrical connection of a semiconductor structure, according to one or more of the embodiments described herein. FIGS. **3**A-**3**C illustrate views of various stages of treating an electrical connection of a semiconductor structure in accordance with one or more embodiments described herein. Although FIGS. **3**A-**3**C are described in relation to the method **200**, the structures disclosed in FIGS. **3**A-**3**C are not limited to the method **200**, but instead may stand alone as structures independent of the method **200**. Similarly, although the method **200** is described in relation to FIGS. **3**A-**3**C, the method **200** is not limited to the structures disclosed in FIGS. **3**A-**3**C but instead may stand alone independent of the structures disclosed in FIGS. **3**A-**3**C. It should be understood that FIGS. **3**A-**3**C illustrate only partial schematic views of the semiconductor device structure **300**, and the semiconductor device structure **300** may contain any number of transistor sections and additional materials having aspects as illustrated in the figures. It should also be noted that although the method **200** illustrated in FIG. **2** is described sequentially, other process sequences that include one or more operations that have been omitted and/or added, and/or has been rearranged in another desirable order, fall within the scope of the embodiments of the disclosure provided herein.

[0026] The term “substrate” as used herein refers to a layer of material that serves as a basis for subsequent processing operations. The substrate may be a silicon based material or any suitable insulating materials or conductive materials as needed. The substrate may include a material such as crystalline silicon (e.g., Si<100> or Si<111>), silicon oxide, strained silicon, silicon germanium, doped or undoped polysilicon, doped or undoped silicon wafers and patterned or non-patterned wafers, silicon on insulator (SOI), carbon doped silicon oxides, silicon nitride, doped silicon, germanium, gallium arsenide, glass, or sapphire.

[0027] Referring to FIG. **2**, at operation **210**, a semiconductor device structure **300** having a feature formed therein is provided. FIG. **3**A illustrates a cross-sectional view of the semiconductor device structure **300** during intermediate stages of manufacturing corresponding to the operation **210**. The semiconductor device structure **300** includes a device substrate **302** having one or more layers formed thereon, for example, a dielectric layer **304** as is shown in FIG. **3**A. The device substrate **302** may be or include a bulk semiconductor substrate, a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type dopant or an n-type dopant) or undoped. In some embodiments, the semiconductor material of the device substrate **302** may include an elemental semiconductor, for example, such as silicon (Si) or germanium (Ge); a compound semiconductor including, for example, silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including, for example, SiGe, GaAsP, AlInAs, GaInAs, GaInP, and/or GaInAsP; a combination thereof, or the like. The device substrate **302** may include additional materials, for example, silicide layers, metal silicide layers, metal layers, dielectric layers, etch stop layers, interlayer dielectrics, or a combination thereof.

[0028] The device substrate **302** may further include integrated circuit devices (not shown). As one of ordinary skill in the art will recognize, a wide variety of integrated circuit devices such as transistors, diodes, capacitors, resistors, the like, or combinations thereof may be formed in and/or on the device substrate **302** to generate the structural and functional requirements of the design for the resulting semiconductor device structure **300**.

[0029] The device substrate **302** has a frontside **302*f*** (also referred to as a front surface) and a backside **302*b*** (also referred to as a back surface) opposite the frontside **302*f***. The dielectric layer **304** is formed over the frontside **302*f*** of the device substrate **302**. The dielectric layer **304** may include multiple layers. The dielectric layer **304** includes an upper surface **304*u*** or field region. In some embodiments, the dielectric layer **304** includes a dielectric material, such as a low k dielectric (SiCOH), silicon oxide, silicon dioxide (SiO.sub.2), silicon nitride (Si.sub.3N.sub.4), silicon carbide (SiC), silicon oxynitride (SiON), aluminum oxide (Al.sub.2O.sub.3), aluminum nitride (AlN), a combination thereof, or multi-layers thereof. In some embodiments, the dielectric layer **304** consists essentially of silicon oxide. It is noted that the foregoing descriptors for example, silicon oxide, should not be interpreted to disclose any particular stoichiometric ratio. Accordingly, “silicon oxide” and the like will be understood by one skilled in the art as a material consisting essentially of silicon and oxygen without disclosing any specific stoichiometric ratio.

[0030] The semiconductor device structure **300** is patterned to form one or more feature(s) **306**. The feature **306** may be a high aspect ratio (HAR) feature. In some embodiments, the feature **306** can be selected from, but not limited to, a trench, a via, a hole, a cavity, or a combination thereof. In particular embodiments, the feature is a trench. In other particular embodiments, the feature **306** is a via. In some embodiments, the feature **306** extends from the upper surface **304*u*** of the dielectric layer **304** towards the backside **302*b*** of the device substrate **302**. The feature **306** includes sidewall surface(s) **306*s*** that extend from the field region **304*u*** to the backside **302*b***.

[0031] In some embodiments, an electrical connection, such as electrical connection **303** is formed within the feature **306**. The electrical connection **303** may be an interconnect, a contact structure, or the like. The electrical connection **303** may be formed by depositing a conductive material into the feature **306**. The conductive material may be deposited such that the conductive material fills the entire or at least a portion of the feature **306**. For example, as shown in FIG. **3**A, the electrical connection **303** may be a contact structure formed by filling a bottom portion **306***b* of the feature **306** with a conductive material. In some embodiments, the bottom portion **306***b* is defined by (extends between) the frontside **302***f* and the backside **302***b* of the device substrate **302**. Therefore, the electrical connection **303** may extend between the frontside **302***f* and the backside **302***b* of the device substrate **302** (or any other portion of the feature **306**, including the entire feature **306**). The conductive material may be formed of copper (Cu), cobalt (Co), molybdenum (Mo), tungsten (W), or ruthenium (Ru). The feature **306** has a first depth “D1” from the upper surface **304***u* to the backside **302***b* and a width “W1” between the two sidewall surface(s) **306***s*. In some embodiments, the depth D1 is in a range of 2 nm to 200 nm. In some embodiments, the width W1 is in a range of 10 nm to 100 nm. In some embodiments, the feature **306** has an aspect ratio (D/W) in a range of 1 to 20.

[0032] In some embodiments, the electrical connection **303** has a high resistivity due to a high amount of contaminants (impurities) that reside in the conductive material. Embodiments herein relate to heating the semiconductor device structure **300** using a microwave process, causing the impurities to rise to the surface of the electrical connection **303**, and then exposing the semiconductor device structure **300** (i.e., the conductive material) to reactive gas to remove the impurities and reduce the resistivity of the electrical connection **303**.

[0033] Referring to FIG. **2**, at operation **220**, the semiconductor device structure **300** is exposed to (i.e., undergoes) a microwave process. FIG. **3**B illustrates a cross-sectional view of a semiconductor device structure **300** during the microwave process corresponding to the operation **220**. The microwave process of operation **220** can include exposing the semiconductor device structure **300** to microwaves in a processing chamber such as the first processing chamber **132** (FIG. **1**). The microwave process of operation **220** can include one or more exposures to microwaves. In one or more examples, the semiconductor device may be exposed to microwaves having a frequency between 300 MHz and 1000 GHz at a power between 100-500 W for a time period between 5-30 min in a processing chamber maintained at a temperature between 300-450° C. and at a pressure between 50-500 mTorr. As shown in FIG. **3**B, during the microwave process, microwaves contact the dielectric layer **304** and the electrical connection **303** and heats the electrical connection **303** more than the dielectric layer **304** because the electrical connection is made from a conductive material. Advantageously the microwaves react with the conductive material and cause a layer of impurities **308** to form on the surface of the electrical connection **303**. Stated differently, the microwaves react with the conductive material and force all the impurities to the surface of the electrical connection **303**. For example, the microwaves penetrate the bulk of the electrical connection **303** and expose a layer of impurities on the surface of the electrical connection **303**. For example, if the conductive material is Ru, the layer of impurities **308** may include, but are not limited to, carbon, iodine or the like. In another example the layer of impurities **308** may include an oxide layer formed on the surface of the electrical connection **303**.

[0034] At operation **230**, as shown in FIG. **3**C, the semiconductor device structure **300** is exposed to a reactive gas in a processing chamber. The exposure to the reactive gas may be performed in a same or different processing chamber than the microwave process. In one or more examples, the reactive gas includes, but is not limited to hydrogen. In one example, the exposure to the reactive gas removes the layer of impurities **308**. Advantageously, the microwave process causes the impurities to rise to the surface of the contact structure without the use of plasma. This reduces damage to the dielectric layer **304**, improves the quality of the conductive material (removes the impurities), exposes the impurities (i.e., layer of impurities **308**) so the impurities are easily removed, removes an oxide layer formed on the surface of the electrical connection **303**, and reduces the resistivity of the electrical connection **303**.

Processing Chamber Example

[0035] Referring now to FIGS. **4**A-**4**D, a series of illustrations depicting an example of a microwave processing tool **400** is shown, in accordance with an embodiment. The microwave processing tool **400** is configured to deliver microwave energy to a processing region of the process chamber to remove impurities from the electrical connection **303** (FIGS. **3**A-**3**C).

[0036] Referring now to FIG. **4**A, a cross-sectional illustration of a microwave processing tool **400** (referred to as processing tool **400** for short) is shown, according to an embodiment. In some embodiments, the processing tool **400** may be a processing tool suitable for any type of processing operation that requires the delivery of microwave energy. In some embodiments, one or more of the chambers **110** and **114**, or even chambers **132-140**, may include the processing tool **400**. The processing tool may emit high-frequency electromagnetic radiation in the form of microwave energy. In some embodiments, “High-frequency” may refer to frequencies between 300 MHz and 1000 GHz.

[0037] Generally, embodiments include a processing tool **400** that includes a chamber **478**. In the processing tool **400**, the chamber **478** may be a vacuum chamber. A vacuum chamber may include a pump (not shown) for removing gases from the chamber to provide the desired vacuum. Additional embodiments may include a chamber **478** that includes one or more gas lines **470** for providing processing gases into the chamber **478** and exhaust lines **472** for removing byproducts from the chamber **478**. While not shown, it is to be appreciated that gas may also be injected into the chamber **478** through a source array **450** (e.g., as a showerhead) for evenly distributing the processing gases over a workpiece **474** (e.g., wafer, substrate, etc.).

[0038] In an embodiment, the workpiece **474** may be supported on a chuck **476**. For example, the chuck **476** may be any suitable chuck, such as an electrostatic chuck. The chuck **476** may also include cooling lines and/or a heater to provide temperature control to the workpiece **474** during processing. Due to the modular configuration of the high-frequency emission modules described herein, embodiments allow for the processing tool **400** to accommodate any sized workpiece **474**. For example, the workpiece **474** may be a semiconductor wafer (e.g., 200 mm, 300 mm, 450 mm, or larger). Alternative embodiments also include workpieces **474** other than semiconductor wafers. For example, embodiments may include a processing tool **400** configured for processing glass substrates, (e.g., for display technologies).

[0039] According to an embodiment, the processing tool **400** includes a modular high-frequency emission source **404**. The modular high-frequency emission source **404** may comprise an array of high-frequency emission modules **405**. In an embodiment, each high-frequency emission module **405** may include an oscillator module **406**, an amplification module **430**, and an applicator **442**. As shown, the applicators **442** are schematically shown as being integrated into the source array **450**

[0040] In an embodiment, the oscillator module **406** and the amplification module **430** may comprise electrical components that are solid state electrical components. In an embodiment, each of the plurality of oscillator modules **406** may be communicatively coupled to different amplification modules **430**. For example, each oscillator module **406** may be electrically coupled to a single amplification module **430**. In an embodiment, the plurality of oscillator modules **406** may generate incoherent electromagnetic radiation. Accordingly, the electromagnetic radiation induced in the chamber **478** will not interact in a manner that results in an undesirable interference pattern.

[0041] In an embodiment, each oscillator module **406** generates high frequency electromagnetic radiation that is transmitted to the amplification module **430**. After processing by the amplification module **430**, the electromagnetic radiation is transmitted to the applicator **442**. In an embodiment, the applicators **442** each emit electromagnetic radiation into the chamber **478**. In some embodiments, the applicators **442** couple the electromagnetic radiation to the processing gases in the chamber **478** to provide energy thereto, without forming a plasma.

[0042] Referring now to FIG. **4**B, a schematic of a solid state high-frequency emission module **405** is shown, in accordance with an embodiment. In an embodiment, the high-frequency emission module **405** comprises an oscillator module **406**. The oscillator module **406** may include a voltage control circuit **410** for providing an input voltage to a voltage controlled oscillator **420** in order to produce high-frequency electromagnetic radiation at a desired frequency. The voltage controlled oscillator **420** is an electronic oscillator whose oscillation frequency is controlled by the input voltage. According to an embodiment, the input voltage from the voltage control circuit **410** results in the voltage controlled oscillator **420** oscillating at a desired frequency.

[0043] According to an embodiment, the electromagnetic radiation is transmitted from the voltage controlled oscillator **420** to an amplification module **430**. The amplification module **430** may include a driver/pre-amplifier **434**, and a main power amplifier **436** that are each coupled to a power supply **439**. According to an embodiment, the amplification module **430** may operate in a pulse mode. For example, the amplification module **430** may have a duty cycle between 1% and 99%. In a more particular embodiment, the amplification module **430** may have a duty cycle between approximately 15% and 50%.

[0044] In an embodiment, the electromagnetic radiation may be transmitted to the thermal break **449** and the applicator **442** after being processed by the amplification module **430**. However, part of the power transmitted to the thermal break **449** may be reflected back due to the mismatch in the output impedance. Accordingly, some embodiments include a detector module **481** that allows for the level of forward power **483** and reflected power **482** to be sensed and fed back to the control circuit module **421**. It is to be appreciated that the detector module **481** may be located at one or more different locations in the system (e.g., between the circulator **438** and the thermal break **449**). In an embodiment, the control circuit module **421** interprets the forward power **483** and the reflected power **482**, and determines the level for the control signal **485** that is communicatively coupled to the oscillator module **406** and the level for the control signal **386** that is communicatively coupled to the amplification module **430**. In an embodiment, control signal **485** adjusts the oscillator module **406** to optimize the high-frequency radiation coupled to the amplification module **430**. In an embodiment, control signal **486** adjusts the amplification module **430** to optimize the output power coupled to the applicator **442** through the thermal break **449**. In an embodiment, the feedback control of the oscillator module **406** and the amplification module **430**, in addition to the tailoring of the impedance matching in the thermal break **449**, may allow for the level of the reflected power to be less than approximately 5% of the forward power. In some embodiments, the feedback control of the oscillator module **406** and the amplification module **430** may allow for the level of the reflected power to be less than approximately 2% of the forward power.

[0045] Referring now to FIG. **4**C, a perspective view illustration of a source array **450** is shown, in accordance with an embodiment. In an embodiment, the source array **450** comprises a dielectric plate **460**. A plurality of cavities **467** are disposed into a first surface **461** of the dielectric plate **460**. The cavities **467** do not pass through to a second surface **462** of the dielectric plate **460**. The source array **450** may further include a plurality of dielectric resonators **466**. Each of the dielectric resonators **466** may be in a different one of the cavities **467**. Each of the dielectric resonators **466** may comprise a hole **465** in the axial center of the dielectric resonator **466**.

[0046] In an embodiment, the dielectric resonators **466** may have a first width W1, and the cavities **467** may have a second width W2. The first width W1 of the dielectric resonator **466** is smaller than the second width W2 of the cavities **467**. The difference in the widths provides a gap G between a sidewall of the dielectric resonators **466** and a sidewall of the cavity **467**. In the illustrated embodiment, each of the dielectric resonators **466** are shown as having a uniform width W1. However, it is to be appreciated that not all dielectric resonators **466** of a source array **450** need to have the same dimensions.

[0047] Referring now to FIG. **4**D, a cross-sectional illustration of a processing tool **400** that includes an assembly **490** is shown, in accordance with an embodiment. In an embodiment, the processing tool comprises a chamber **478** that is sealed by an assembly **490**. For example, the assembly **490** may rest against one or more O-rings **495** to provide a vacuum seal to a chamber volume **493** of the chamber **478**. In other embodiments, the assembly **490** may interface with the

chamber **478**. That is, the assembly **490** may be part of a lid that seals the chamber **478**. In an embodiment, the processing tool **400** may comprise a plurality of processing volumes (which may be fluidically coupled together), with each processing volume having a different assembly **490**. In an embodiment, a chuck **476** or the like may support a workpiece **474**. The workpiece **474** may be a distance D from the assembly **490**. In an embodiment, the chamber volume **493** may be suitable for delivering microwave energy to a process gas disposed within the chamber **478**. That is, the chamber **478** may be a vacuum chamber.
[0048] In an embodiment, the assembly **490** comprises a source array **450** and a housing **492**. The source array **450** may comprise a dielectric plate **460** and a plurality of dielectric resonators **466** extending up from the dielectric plate **460**. Cavities **467** into the dielectric plate **460** may surround each of the dielectric resonators **466**. Sidewalls of the cavity **467** are separated from the sidewall of the dielectric resonator **466** by a gap G. The dielectric plate **460** and the dielectric resonators **466** of the source array **450** may be a monolithic structure (as shown in FIG. **4**D), or the dielectric plate **460** and the dielectric resonators **466** may be discrete components.
[0049] The housing **492** include rings **431** that fit into the gaps G. In an embodiment, the rings **431** and the conductive body **473** of the housing **492** are a monolithic structure (as shown in FIG. **4**D), or the conductive body **473** and the rings **431** may be discrete components. The housing **492** may having openings sized to receive the dielectric resonators **466**. In an embodiment, monopole antennas **488** may extend into holes in the dielectric resonators **466**. The monopole antennas **488** are each electrically coupled to power sources (e.g., high-frequency emission modules **405**).
[0050] While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A method comprising: exposing a semiconductor device structure to a microwave process to cause impurities within at least one electrical connection formed in at least one feature of the semiconductor device structure to rise to a surface of the at least one electrical connection; and exposing the semiconductor device structure to a reactive gas to remove the impurities from the surface of the at least one electrical connection.
2. The method of claim 1, wherein the at least one electrical connection comprises a conductive material deposited within the at least one feature.
3. The method of claim 2, wherein the conductive material comprises ruthenium (Ru).
4. The method of claim 3, wherein the impurities comprise carbon (C), iodine (I), and combinations thereof.
5. The method of claim 1, wherein the reactive gas comprises hydrogen.
6. The method of claim 1, wherein the impurities further comprise an oxide layer formed on the surface of the electrical connection.
7. The method of claim 1, wherein the semiconductor device structure comprises a dielectric layer formed on a frontside of a device substrate and the semiconductor device structure is patterned to form the at least one feature, wherein the at least one feature extends between a field region of the dielectric layer to a backside of the device substrate.
8. The method of claim 7, wherein the at least one electrical connection is formed within the at least one feature and extends between the frontside of the device substrate and the backside of the device substrate.
9. The method of claim 1, wherein the exposing a semiconductor device structure to a microwave process and the exposing the semiconductor device structure to a reactive gas are performed in a same processing chamber.
10. The method of claim 1, wherein the exposing a semiconductor device structure to a microwave

process and the exposing the semiconductor device structure to a reactive gas are performed in different processing chambers.

11. A processing tool comprising: a controller; and a memory for storing instructions, which, when executed by the controller, causes the controller to perform a method for treating an electrical connection formed in a feature of a semiconductor structure, the method comprising: exposing a semiconductor device structure to a microwave process to cause impurities within at least one electrical connection formed in at least one feature of the semiconductor device structure to rise to a surface of the at least one electrical connection; and exposing the semiconductor device structure to a reactive gas to remove the impurities from the surface of the at least one electrical connection.

12. The processing tool of claim 11, wherein the processing tool further comprises a first processing chamber configured to perform the exposing a semiconductor device structure to a microwave process and a second processing chamber configured to perform the exposing the semiconductor device structure to a reactive gas to remove the impurities from the surface of the at least one electrical connection.

13. The processing tool of claim 11, wherein the at least one electrical connection comprises a conductive material deposited within the at least one feature.

14. The processing tool of claim 13, wherein the conductive material comprises ruthenium (Ru).

15. The processing tool of claim 14, wherein the impurities comprise carbon (C), iodine (I), and combinations thereof.

16. The processing tool of claim 11, wherein the reactive gas comprises hydrogen.

17. The processing tool of claim 11, wherein the impurities further comprise an oxide layer formed on the surface of the electrical connection.

18. The processing tool of claim 11, wherein the semiconductor device structure comprises a dielectric layer formed on a frontside of a device substrate and the semiconductor device structure is patterned to form the at least one feature, wherein the at least one feature extends between a field region of the dielectric layer to a backside of the device substrate.

19. The processing tool of claim 18, wherein the at least one electrical connection is formed within the at least one feature and extends between the frontside of the device substrate and the backside of the device substrate.

20. A method comprising: exposing a semiconductor device structure to a microwave process to cause impurities in an electrical connection comprising ruthenium (Ru) that is formed in a feature of the semiconductor device structure to rise to a surface of electrical connection; and exposing the semiconductor device structure to hydrogen to remove the impurities from the surface of the electrical connection.
